

Karnataka Joint Gravity–Magnetic Inversion: Couplings and Remanence

The gravity and the magnetic data of the same area inverted together on one mesh with terrain, once for each coupling of the joint inversion, and the magnetic data alone with a magnetization vector per cell. The report shows what each coupling does to the two models, what it costs in data fit, which assumption suits this belt, and what the magnetic data say about the direction of magnetization.

AREA	DATA
70 × 70 km, easting 641–711 km, northing 1634–1704 km (UTM 43N, EPSG:32643)	Complete Bouguer anomaly, 5,040 points, error 0.5 mGal; magnetic anomaly continued to 1 km above the ground, 5,041 points, error 5% + 10 nT; second-order trends removed
MODELS	RUNS
Density contrast (−0.2 to +0.5 g/cc) and susceptibility (0 to 1 SI) on 335,518 cells below the ground (1 km × 1 km × 250 m); each sparse with $\alpha_s = 1$ and depth weighting $\beta = 1$	6 joint inversions and one magnetization-vector inversion on AWS EC2 (ap-south-1, c5.9xlarge and c5.18xlarge), 22–124 minutes each, \$13.62 in total; the same couplings on a 2 km mesh locally

Summary

- **Without a coupling the joint inversion returns the two single inversions** (cell-by-cell correlation 0.95 and 0.76). The two uncoupled models have little in common cell by cell: 21% of the magnetic cells are dense or light, and 32% of the susceptibility model's edges lie on an edge of the density model.
- **The two structural couplings act on the susceptibility model and leave the density model alone** (correlation with the uncoupled density 0.92 and 0.97; susceptibility 0.60 and 0.61). The cross-gradient makes the gradients parallel (measure 0.44 → 0.07), but mostly by keeping the edges of the two models apart (2% shared); the joint total variation puts more of them in the same cells (47%), with both datasets fitted (χ^2/N 1.09 and 1.03).
- **The linear correspondence forces one relation on the belt and spoils the density model:** the dense body under the main high moves to 0.0–3.8 km (uncoupled 3.0–7.5 km) and 45% of the density model ends below the core, while both datasets are still fitted (χ^2/N 1.04 and 1.04).
- **The group lasso fits both datasets and keeps its models in the core** (χ^2/N 1.04 and 0.97; 89% and 75% in the core), **and it pairs the two supports:** 81% of its magnetic cells are anomalous in density (21% uncoupled, 30% in its control), and it lines up the two structures (75% of the susceptibility edges on a density edge, cross-gradient measure 0.13). Both models move: the dense body under the main high rises to 2.0–5.2 km (uncoupled 3.0–7.5 km), and the susceptibility spreads through the dense body.
- **The magnetic data ask for remanent magnetization.** With a magnetization vector per cell the RMS residual falls from 49 to 28 nT, and at the 92 stations south of the Sandur belt that every susceptibility model

underfits by more than 150 nT from 321 to 129 nT (RMS; 22 of the 92 are still off by more than 150 nT); the strongly magnetized cells point at I 74°, D -79°, not along the present field (I 19°, D -1°).

- **For this belt the structural assumption is the right one.** The magnetic rock (iron formation) lies on the margins of the dense rock (the greenstone pile), so the two models share boundaries, not values: a coupling that ties the values together, by one relation or by one support, moves one model to where the other is. The data fit does not tell the couplings apart, so the choice has to come from the geology.
- **No coupling fixes the depth.** The depth of each model still follows its own regularization (β); a coupling moves one model towards the other, which helps only where they really are the same rock.

1 Set-up

The two datasets are those of the single-method reports: the Bouguer anomaly with the terrain correction at 1 km, and the magnetic anomaly continued upwards to 1 km above the ground. They share one mesh that follows the ground (84 × 84 × 50 cells, 335,518 below the ground), so that a cell of the density model and a cell of the susceptibility model are the same rock.

In the runs of the SimPEG couplings each model is regularized as in the reference run of its own report: sparse (IRLS) with $\alpha_s = 1$, $p = [0, 2, 2, 2]$ and Li & Oldenburg depth weighting with $\beta = 1$, within its bounds. One trade-off parameter serves both models; the pipeline balances the two regularizations against their data and cools it until $\chi^2 = N$ for the 10,081 data together, with at most 60 iterations and 40 IRLS iterations. The coupling weight is 1 in the pipeline's unit-free scale (as strong as the regularization).

The group lasso has its own regularization: an L2 term ($\lambda_2 = 0.3$) and the group norm of each cell's (density, susceptibility) pair, solved by ADMM within the same bounds. Its cells are weighed like those of the other runs (cell volume × the depth weight with $\beta = 1$), each datum by its error, λ_1 is chosen on a sweep of 8 values for $\chi^2 = N$, and the two datasets are then reweighted until each fits its own $\chi^2 = N$. Its control runs the same solver without the pairing: each model soft-thresholded on its own.

Coupling	Kind	What it assumes	Iterations	EC2 machine	Time (min)
No coupling	—	Nothing: the two models only share the mesh.	44	c5.9xlarge	22
Cross-gradient	structural	The gradients of the two models are parallel, or one of them is zero (Gallardo & Meju 2003).	45	c5.9xlarge	26
Joint total variation	structural	The two models change in the same places (Haber & Holtzman Gazit 2013).	44	c5.9xlarge	28
Linear correspondence	petrophysical	One linear relation in every cell: density contrast = 0.5 × susceptibility, the ratio of the two upper bounds.	52	c5.9xlarge	26
Group lasso	joint sparsity	Few cells are anomalous, and a cell is anomalous in both models or in neither (Utsugi 2025).	8 values of λ_1	c5.18xlarge	124
L1 + L2 by ADMM, no coupling (the group lasso's control)	control	Few cells are anomalous in each model, no pairing: what the group lasso's solver does without the coupling.	8 values of λ_1	c5.18xlarge	66

The couplings are those of `geoinv3d/methods/coupling.py` (docs/joint_couplings.md); the group lasso of `geoinv3d/methods/group_lasso.py` (docs/group_lasso_joint.md).

2 Results

2.1 Data fit

Coupling	Gravity χ^2/N	RMS (mGal)	Max residual (mGal)	Magnetic χ^2/N	RMS (nT)	Max residual (nT)
Single inversions (same settings)	0.93			0.90		
No coupling	0.95	0.49	4.9	0.95	47	585
Cross-gradient	1.10	0.53	4.9	0.88	53	912
Joint total variation	1.09	0.52	5.1	1.03	53	729
Linear correspondence	1.04	0.51	4.7	1.04	42	568
Group lasso	1.04	0.51	5.8	0.97	55	844
L1 + L2 by ADMM, no coupling (the group lasso's control)	1.05	0.51	5.7	0.96	55	815

Red: χ^2/N outside 0.7–1.3. The first row gives the single inversions with the same settings.

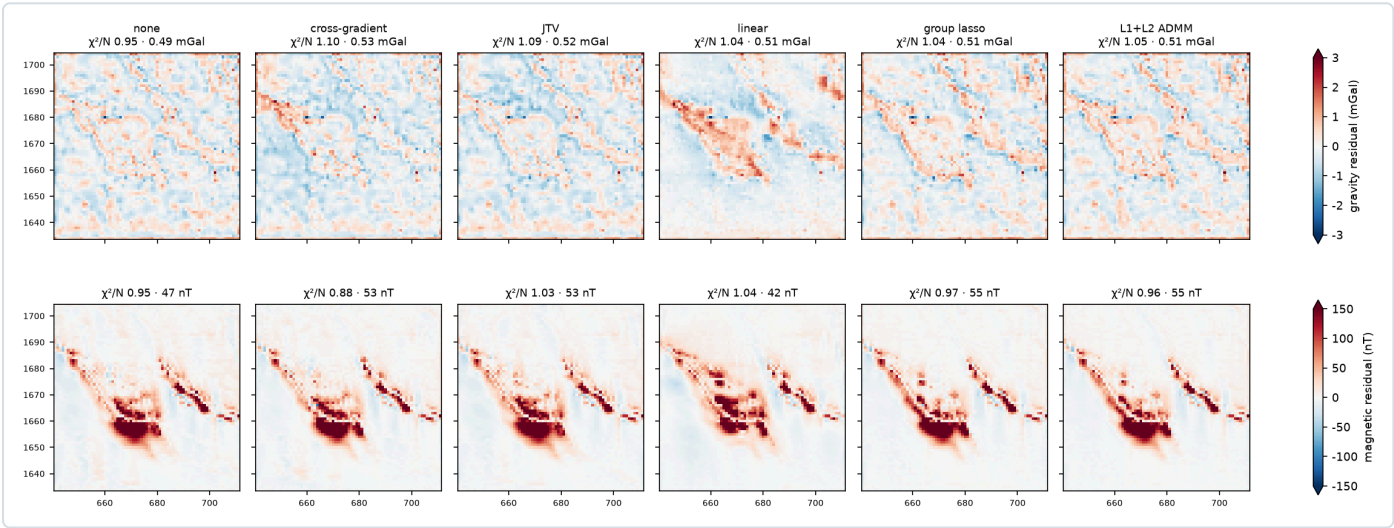


Figure 1 Residuals (observed – predicted) of the gravity data (top, ± 3 mGal) and the magnetic data (bottom, ± 150 nT) for each coupling.

Every run fits both datasets to their errors: χ^2/N lies between 0.88 and 1.10 for gravity and magnetics alike. The data fit therefore does not rank the couplings; what they do differently shows in the models (Sections 2.2 and 2.3).

Inverted together without a coupling, the two models are those of the single inversions (cell-by-cell correlation 0.95 for density and 0.76 for susceptibility): the joint set-up itself changes nothing. The positive magnetic residual south of the Sandur belt is there in every run: no coupling repairs what a magnetization along the present field cannot explain (Section 3).

2.2 What each coupling does to the models

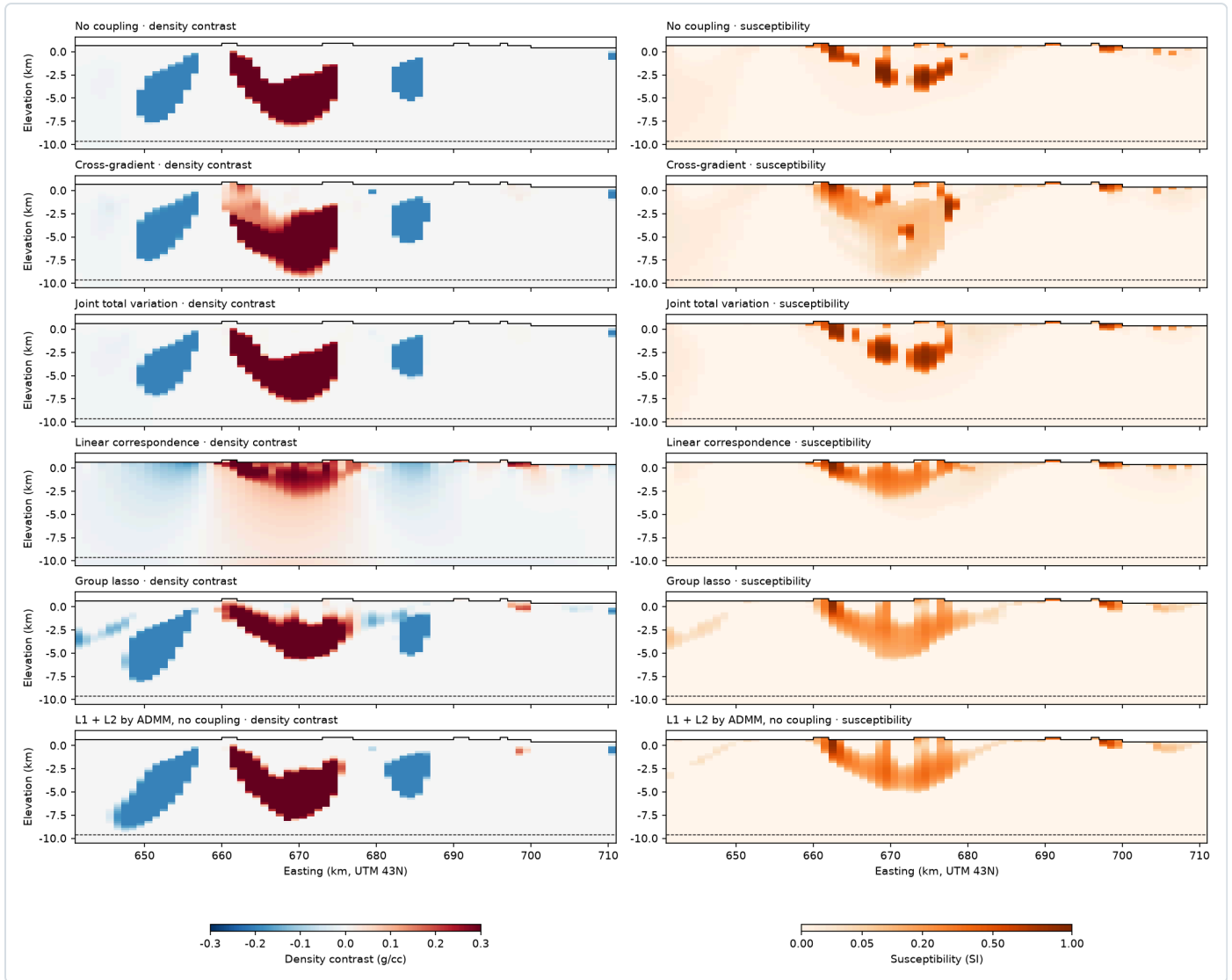


Figure 2 E–W sections through the Sandur belt (northing 1667.5 km): density contrast (left, ± 0.3 g/cc) and susceptibility (right, square-root scale to 1 SI) for each coupling. The black line is the ground of the mesh, the dashed line the base of the core.

Coupling	Density: core	below the core	main high, half-max km	centroid km	correlation with the uncoupled model	Susceptibility: core	below the core	Sandur belt, centroid km	SI·km ³	correlation with the uncoupled model
No coupling	89%	9%	3.0–7.5	5.3	1.00	73%	7%	2.0	150	1.00
Cross-gradient	90%	9%	2.5–8.5	5.6	0.92	70%	8%	2.7	165	0.60
Joint total variation	88%	10%	3.0–7.2	5.2	0.97	73%	10%	2.1	147	0.61
Linear correspondence	41%	45%	0.0–3.8	4.9	0.37	55%	19%	1.4	153	0.43
Group lasso	89%	3%	2.0–5.2	3.7	0.74	75%	6%	2.5	157	0.58
L1 + L2 by ADMM, no coupling (the group lasso's control)	92%	6%	3.2–7.5	5.4	0.92	74%	6%	2.3	152	0.61

Depths below the ground. Main high: the density profile under the main Bouguer high (671.5, 1664.5 km). Sandur belt: susceptibility $\times$ volume in the box of the magnetic report. Correlations are cell by cell in the core columns. Red: less than half in the core, or more than 30% below it.

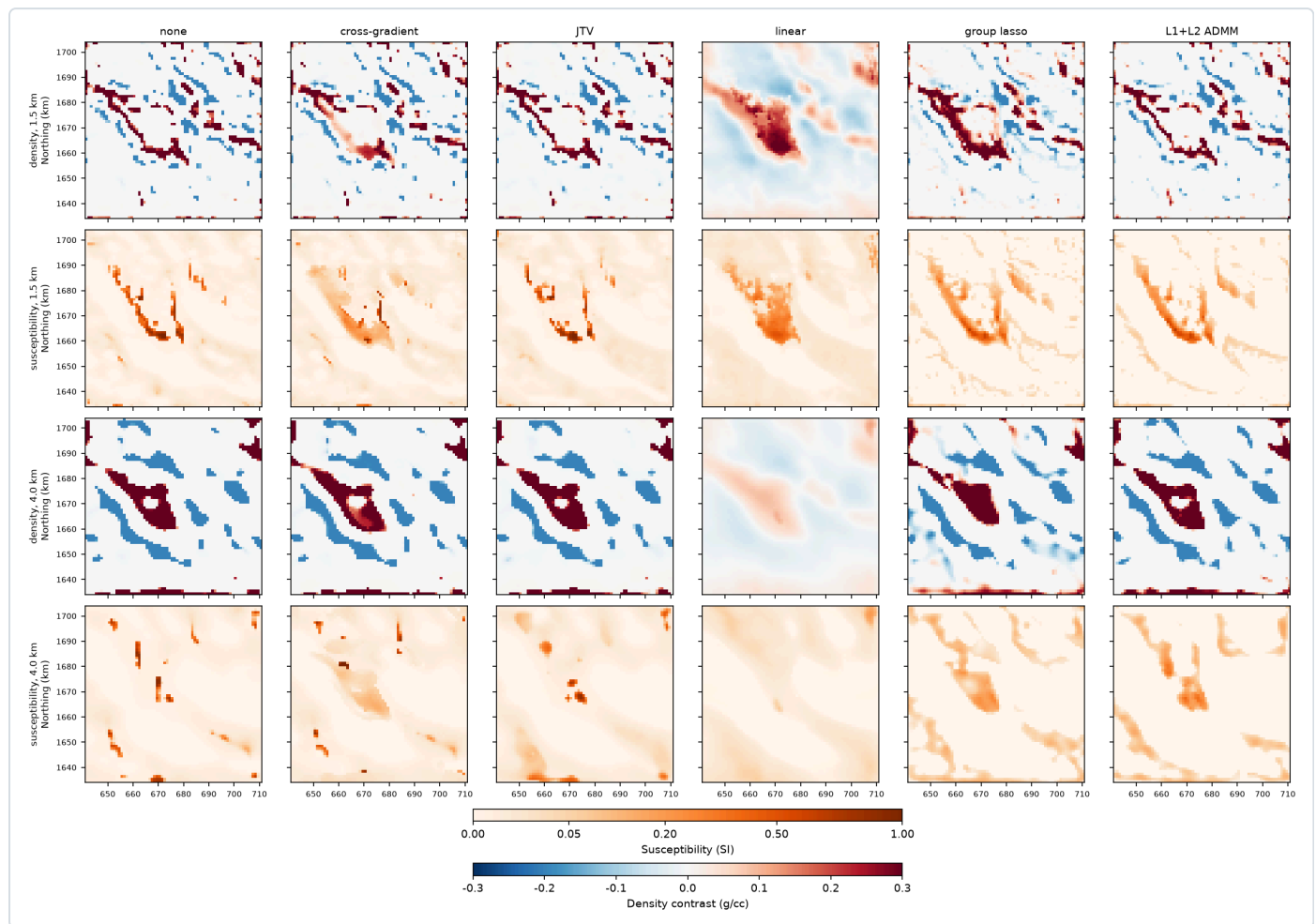


Figure 3 Layers at about 1.5 km and 4 km below the mean ground: density contrast and susceptibility for each coupling (core area only).

- **Cross-gradient.** It does what it is built for: the cross-gradient measure of Section 2.3 falls from 0.44 to 0.07, with both datasets fitted (χ^2/N 1.10 and 0.88). It gets there mostly by the second way of making a cross product zero: only 2% of the susceptibility model's edges lie on an edge of the density model (32% uncoupled), so where one model has an edge the other is flat. The density model hardly changes (correlation 0.92 with the uncoupled one); the susceptibility model changes more (0.60), and its centroid in the belt moves from 2.0 to 2.7 km.
- **Joint total variation.** It rewards edges in the same cells, and the share of susceptibility edges on a density edge rises from 32% to 47%; the directions of the gradients are not constrained (measure 0.39). 38% of the magnetic cells are anomalous in density (21% uncoupled). The density model stays as it was (0.97); both datasets are fitted (χ^2/N 1.09 and 1.03) in 44 iterations.
- **Linear correspondence.** The two models become one: 93% of the magnetic cells are anomalous in density too, and $|\text{density}|$ correlates with susceptibility at 0.71. The relation is wrong for this belt and the density model pays for it: the dense body under the main high has its half-maximum at 0.0–3.8 km, against 3.0–7.5 km uncoupled, and what the relation cannot place is put below the core (45% of the density model, against 9%). Both datasets are still fitted, to χ^2/N 1.04 and 1.04: the fit does not warn.
- **Group lasso.** Both datasets are fitted (χ^2/N 1.04 and 0.97), after 1 round of reweighting the datasets (weights 0.62 for gravity and 1.60 for magnetics: along the λ_1 sweep the gravity data are fitted sooner), and the models stay where the data resolve them (89% of the density and 75% of the susceptibility model in the core). The pairing shows: 81% of the magnetic cells are anomalous in density, against 30% in its control without the pairing

and 21% in the uncoupled SimPEG run, and $|\text{density}|$ correlates with susceptibility at 0.52 (0.06 uncoupled). It also makes the two structures alike: 75% of the susceptibility edges lie on a density edge (32% uncoupled), more than with any coupling but the linear correspondence, and the cross-gradient measure falls to 0.13 (0.44 uncoupled; 0.07 for the cross-gradient itself). The pairing reshapes the density model too (correlation 0.74 with the uncoupled one): the dense body under the main high rises to a half-maximum at 2.0–5.2 km (uncoupled 3.0–7.5 km; its control 3.2–7.5 km; centroid 3.7 km against 5.3 km), towards the magnetic rock, and 3% of the density model lies below the core (9% uncoupled). The susceptibility model spreads through the dense body (correlation 0.58 with the uncoupled one; centroid in the belt 2.5 km, against 2.0 km).

- The group lasso's control** (the same solver, weighting and balance, each model on its own) fits the data to χ^2/N 1.05 and 0.96 and keeps 92% and 74% of its models in the core; 30% of its magnetic cells are anomalous in density. What the group lasso adds to it is the pairing of the two supports.

2.3 How alike the two models are

Measured in the core of the mesh. The cross-gradient measure is the sum of $|\nabla\rho \times \nabla\chi|^2$ over the sum of $|\nabla\rho|^2|\nabla\chi|^2$: 0 when the gradients are parallel wherever both exist, 1 when they are perpendicular. An edge is a cell whose squared gradient exceeds 1% of the model's largest; a cell is anomalous when its value exceeds a tenth of the model's largest.

Coupling	Cross-gradient (0 = parallel gradients)	Susceptibility edges on a density edge	Magnetic cells that are dense or light	Cell-by-cell correlation of $ \text{density} $ and susceptibility	Correlation of the integrated maps
No coupling	0.44	32%	21%	0.06	0.20
Cross-gradient	0.07	2%	31%	0.06	0.26
Joint total variation	0.39	47%	38%	0.13	0.32
Linear correspondence	0.03	90%	93%	0.71	0.59
Group lasso	0.13	75%	81%	0.52	0.39
L1 + L2 by ADMM, no coupling (the group lasso's control)	0.37	39%	30%	0.13	0.33

Green: clearly more alike than the uncoupled models (the first row).

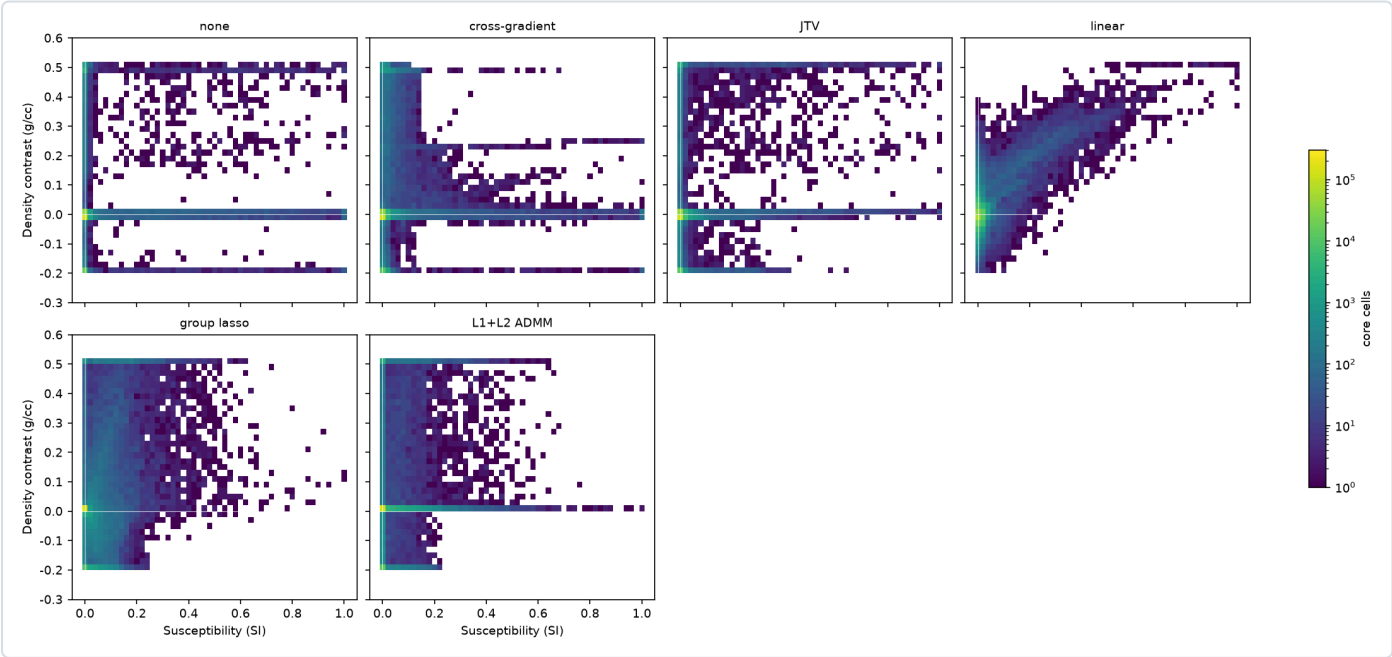


Figure 4 Density contrast against susceptibility, cell by cell in the core (logarithmic counts). Each coupling leaves its own signature.

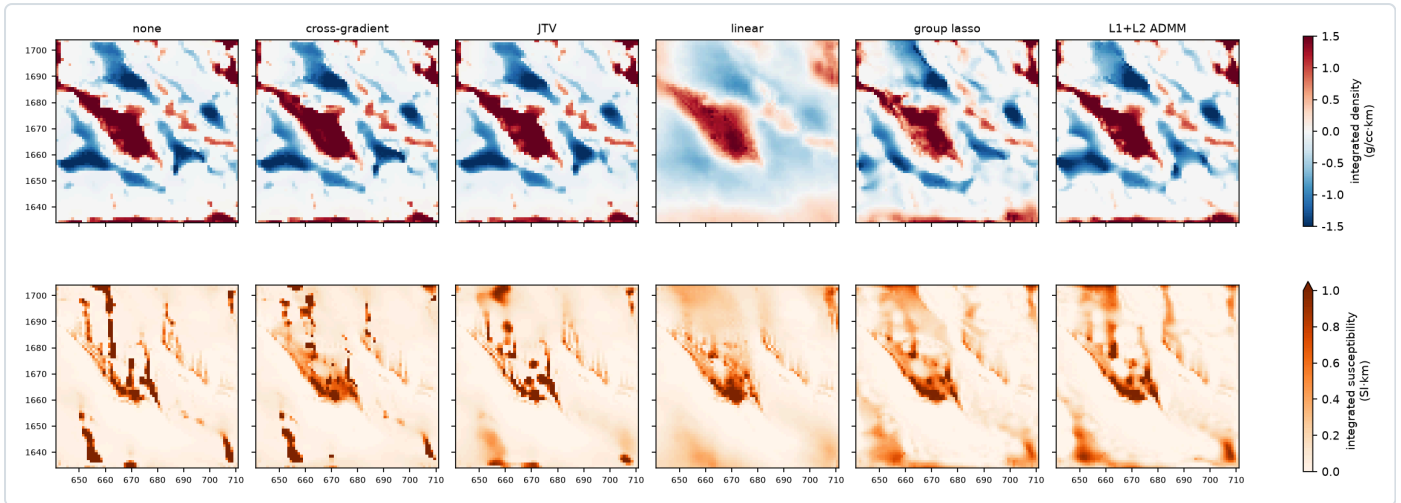


Figure 5 Vertically integrated density (top) and susceptibility (bottom) for each coupling.

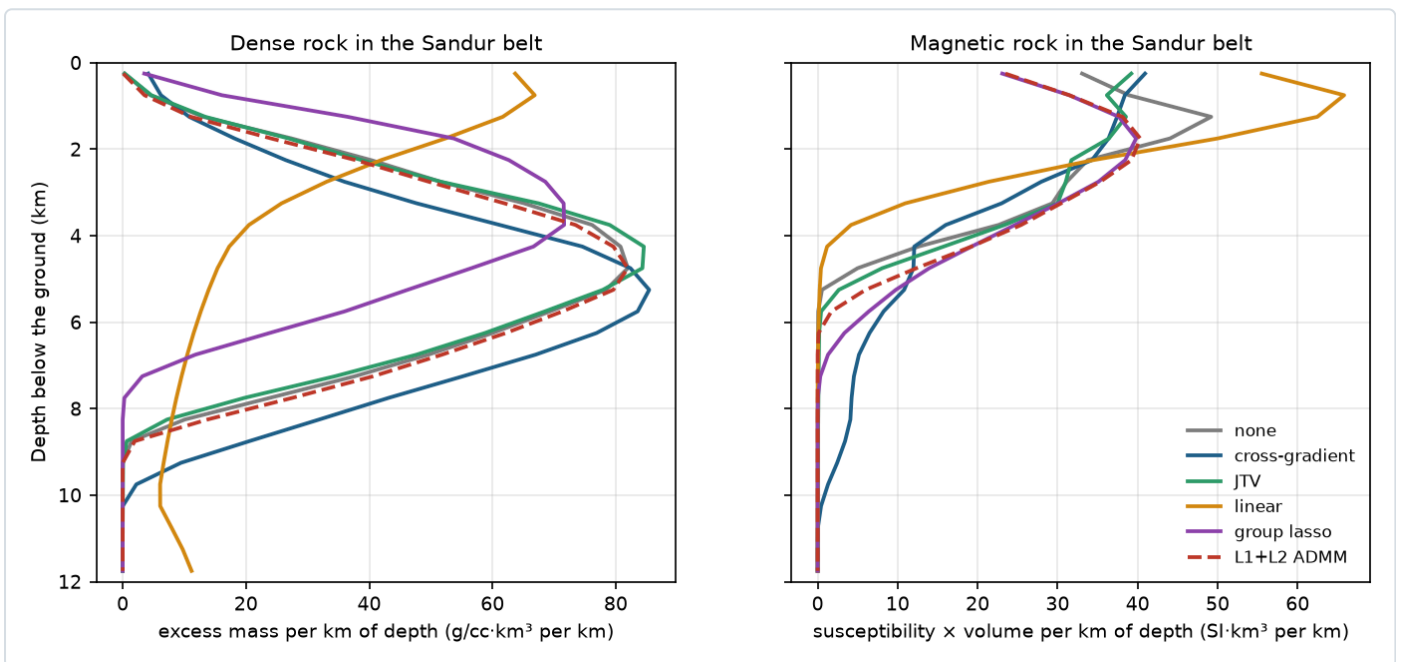


Figure 6 Dense rock (left: positive density contrast $\times$ volume) and magnetic rock (right: susceptibility $\times$ volume) in the Sandur belt, per kilometre of depth below the ground, for each coupling.

3 The magnetic residual: remanent magnetization

A susceptibility model magnetizes every cell along the present field, and a body so magnetized gives a fixed shape to its anomaly: the ratio of its high to its low, and where they lie. The positive anomaly on the southern side of the Sandur belt is fitted worse than the rest: the single magnetic inversion underfits 92 stations by more than 150 nT (easting 664–707 km, northing 1655–1673 km; RMS 321 nT there), and every joint run leaves 88–107 such stations, with an RMS of 224–360 nT at the same 92.

The magnetic data were therefore inverted once more with a magnetization vector in every cell (SimPEG's magnetization-vector inversion: three components of effective susceptibility per cell, a sparse regularization of their amplitude with the same norms, depth weighting and bounds, $|m_i| \leq 1$ SI), which allows a magnetization in any direction. It fits the data to χ^2/N 1.05 with an RMS of 28 nT, against 49 nT for the susceptibility model, and its largest residual is 309 nT against 680 nT. At the 92 underfitted stations the RMS residual falls from 321 to 129 nT (mean $|\text{residual}|$ 293 $\rightarrow$ 105 nT); 22 of them are still off by more than 150 nT (16 above, 6 below the model), so the vector model explains most, not all, of the anomaly.

The strongly magnetized cells (the 1,203 above a tenth of the largest amplitude) point at an inclination of 74° and a declination of -79° (their resultant; coherence 0.75, where 1 means all parallel), far from the present field (I 19°, D -1°). The data ask for a magnetization that is not along the present field: remanent magnetization, common in banded iron formation. The amplitude model is compact: 79% of it lies in the core (71% of the susceptibility model).

For the joint inversions this means that the susceptibility models of the belt are an approximation where the rock is remanent: a coupling can move the susceptibility towards the density model, but it cannot give it the direction the data ask for. The couplings of this report act on the induced susceptibility.

Magnetic inversion ($\beta = 1$)	χ^2/N	RMS (nT)	Max residual (nT)	In the core	Below the core	Sandur belt, centroid km	EC2 machine	Time (min)
Susceptibility along the present field	0.90	49	680	71%	10%	2.1	c5.4xlarge	14
Magnetization vector (MVI)	1.05	28	309	79%	10%	2.1	c5.9xlarge	27

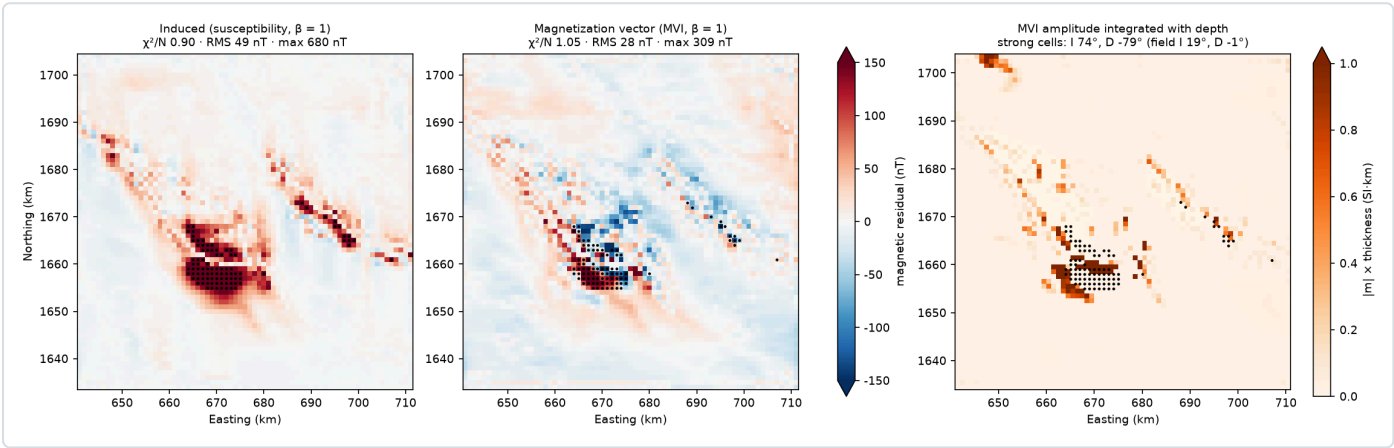


Figure 7 Magnetic residuals of the susceptibility inversion and of the magnetization-vector inversion (± 150 nT; black dots: the stations the susceptibility model underfits by more than 150 nT), and the amplitude of the magnetization vector integrated with depth.

4 Which coupling suits this belt

The single-method reports showed what the two datasets see in the Sandur belt: a thick body of dense, weakly magnetic rock (the greenstone pile; amphibolite and metabasalt samples of 2.9–3.0 g/cc and 0.0002–0.004 SI), and thin sheets of strongly magnetic rock along its margins and its south-eastern closure (the banded iron formation), which are dense too (3.4 g/cc) but too thin to dominate the gravity. A coupling is an assumption about how the two properties are related; on this belt the assumptions fare as follows.

- **Same boundaries (structural couplings): consistent with the geology.** The iron formation bounds the greenstone, so edges of the susceptibility model should lie on edges of the density model. The joint total variation asks for exactly that and delivers it in part; the cross-gradient asks only for parallel gradients and is satisfied, more cheaply, by a smooth susceptibility model whose edges avoid those of the density model. Both leave the density model unchanged: the gravity data and their regularization determine it, and the magnetic sheets are too thin to reshape it.
- **One relation between the values (linear correspondence): contradicted.** The densest rock here is not the most magnetic. Forcing density = 0.5 × susceptibility moves the dense body into the magnetic sheets and leaves the rest of the gravity anomaly to the padding. A relation fitted to samples would fail in the same way, because two rock types with different relations make up the belt.
- **The same cells are anomalous (group lasso): partly true.** The iron formation is both dense and magnetic, the greenstone is dense only. The group lasso lets a cell be anomalous in one property and not the other, at a price, so it does not force the greenstone to be magnetic; but it pulls the magnetic rock into the dense cells (81% of the

magnetic cells are anomalous in density, against 21% uncoupled), and it lifts the dense body towards the magnetic rock (half-maximum 2.0–5.2 km under the main high, against 3.0–7.5 km), as the linear correspondence does, only less. Where a magnetic body is also a dense body, as in the synthetic test of 30 September, this is the assumption to make; here most of the dense rock is not magnetic, and the pairing moves magnetic rock to where there is none.

- **Remanence.** The strongest magnetic rock of the belt is not magnetized along the present field (Section 3). Every coupling here acts on the induced susceptibility, so in the south of the belt the coupled susceptibility models inherit an approximation that no coupling can remove.

The common point: a coupling improves a model where its assumption holds and damages it where it does not, and the data fit says little about which is the case. The measures of Section 2.3 say what a coupling did, not whether it was right.

4.1 Why the couplings that tie the values lift the dense body

The two datasets do not see depth alike. Per unit volume, the magnetic data lose sight of a cell much faster than the gravity data: at 4–8 km below the ground a cell counts 0.22 of what it counts at the surface for gravity but 0.10 for the magnetic data, and below the core 0.11 against 0.027 (the field of a magnetic dipole falls off one power of the distance faster than that of a mass). The magnetic data therefore fix where the magnetic rock is, near the surface, much more firmly than the gravity data fix the depth of the dense rock.

Linear correspondence. With density = 0.5 × susceptibility in every cell, a cell can be dense only if it is magnetic too: the relation asks for 2 SI of susceptibility per g/cc of density, 0.6 SI for a contrast of 0.3 g/cc. The dense body at 3.0–7.5 km of the uncoupled run would need that much susceptibility at depth, whose magnetic anomaly the data do not have; so density is allowed only where the magnetic rock is — the shallow sheets, which carry the dense body up to 0.0–3.8 km under the main high — and where the magnetic data can no longer object: below the core, which receives 45% of the density model (9% uncoupled). The susceptibility model gets shallower too (centroid in the belt 1.4 km, against 2.0 km uncoupled).

Group lasso. The group norm of a cell, $\|(p, \chi)\|$, is less than $|p| + |\chi|$: density costs less in a cell that already holds susceptibility than in an empty one. So the dense rock is drawn towards the shallow magnetic cells, but, with no fixed ratio, it may still stay in non-magnetic cells at the full price: the dense body rises only part of the way, to 2.0–5.2 km, and little goes below the core (3%). Its control, the same solver without the pairing, keeps the dense body at 3.2–7.5 km: the lift comes from the pairing.

The couplings that leave the values free (the cross-gradient and the joint total variation) place no such condition on a cell, and the dense body stays where the gravity data and its own regularization put it. None of the depths is confirmed by the data — every run fits them equally — so the lift shows what the assumption does, not where the rock is.

Depth below the ground (km)	Gravity	Magnetics	Magnetics / gravity
0–2	1.00	1.000	1.00
2–4	0.43	0.295	0.68
4–8	0.22	0.099	0.45
8–10	0.15	0.048	0.33
below 10 (below the core)	0.11	0.027	0.26

Median column norm of the error-weighted sensitivities per km³ of cell, in the core columns of the 2 km mesh, relative to the cells 0–2 km below the ground.

5 The 2 km study

Every coupling was also run locally on a 2 km × 2 km × 500 m mesh (1,296 + 1,296 data):

Run (2 km mesh, local)	Gravity χ^2/N	Magnetic χ^2/N	Density range (g/cc)	Susceptibility range (SI)	Density in the core	Susceptibility in the core	Time (min)
No coupling	0.90	0.90	-0.20 to 0.50	0.00 to 1.00	88%	62%	0.7
Cross-gradient	0.93	0.80	-0.20 to 0.50	0.00 to 1.00	89%	57%	0.8
Joint total variation	0.90	0.91	-0.20 to 0.50	0.00 to 1.00	88%	69%	0.9
Linear correspondence	1.04	0.90	-0.20 to 0.50	0.00 to 0.98	54%	50%	0.8
Group lasso	0.96	1.02	-0.20 to 0.48	0.00 to 0.67	82%	62%	2.5
L1 + L2 by ADMM, no coupling (the group lasso's control)	0.97	1.02	-0.20 to 0.50	0.00 to 0.66	87%	61%	1.0

Red: χ^2/N outside 0.7–1.3.

- **The coarse runs behave like the full-resolution ones:** the cross-gradient lowers the cross-gradient measure, the joint total variation raises the share of common edges, the linear correspondence ties the values, and the group lasso pairs the supports. Each coupling took at most 2.5 minutes on a laptop (group lasso): enough to choose a coupling before paying for the full runs.
- **The group lasso on the 2 km mesh** fits the data to χ^2/N 0.96 and 1.02 and keeps 82% and 62% of its models in the core (89% and 75% at full resolution); 74% of its magnetic cells are anomalous in density (81% at full resolution).

6 Conclusions and recommendations

- **Choose the coupling from the geology, not from the fit.** Same rock in both models (one body, dense and magnetic): the group lasso. Different rock sharing boundaries, as in this belt: a structural coupling. Unknown: invert separately first and compare the two models.
- **For this area: the joint total variation or the cross-gradient,** read together with the uncoupled models. Report what is common to them; do not read the coupled susceptibility model as better resolved.
- **Use the group lasso where the magnetic bodies are also the dense ones** (the iron formation alone, or a mafic intrusion); over the whole belt it draws magnetic rock into the greenstone.
- **Do not use the linear correspondence** unless one rock type carries both anomalies.
- **Invert the magnetic data of the belt for a magnetization vector.** The susceptibility models are an approximation where the iron formation is remanent; a joint inversion that couples the amplitude of the magnetization vector, rather than the susceptibility, would carry the remanence into the joint model. Oriented samples of the iron formation would tell whether the direction found (Section 3) is its remanence.
- **Depth still needs independent information.** The couplings do not supply it; the mapped dips of the iron formation or boreholes would.

Files

- Parameters of every run: `examples/output/karnataka_joint/scripts/joint_params.py` (joint) and `deploy/ec2_multi_run.py` (the magnetic runs `beta1` and `beta1_mvi`); launched with `deploy/ec2_multi_run.py` and `scripts/run_lowres.py --out data/lowres_runs_fixed`
- This report and its figures: `scripts/make_figures.py --set v2`, `scripts/build_report_v2.py`, `scripts/report_text_v2.py`; results in `data/ec2_runs_fixed/`, `data/lowres_runs_fixed/` and `karnataka_magnetic/data/ec2_runs/`
- The single-method reports: `examples/output/karnataka_gravity_terrain/`, `examples/output/karnataka_magnetic/`